

2020 Mathematics Standard Year 11 – Examination Worked Solutions

1. B

As Vanessa paid 60% for the dress, she must have paid 40% for the shoes.

$$0.40 \times \$350 = \$140$$

2. D

Selecting every 20th TV is an example of a systematic sampling process.

3. C

For Saturday Kevin is paid \$ 1 418 – \$ 1 152 = \$ 266.

The number of hours worked is \$ 266 ÷ \$ 38 = 7

Hence, Kevin worked for 7 hours on Saturday.

4. A

Melissa's profit was \$ 525 – \$ 500 = \$ 25

Hence, her percentage profit is

$$\frac{25}{500} \times 100 = 5\%$$

5. C

The team has an average of 4 goals in 12 games, so, the total number of goals in these games is

$$12 \times 4 = 48. \text{ Hence, the team scored}$$

$$48 - 30 = 18 \text{ goals in the last 3 games.}$$

Alternative method

Let the number of goals in the last 3 games be x .

$$\text{So } \frac{30+x}{12} = 4 \text{ then } 30 + x = 48 \text{ that is } x = 18.$$

6. B

The volume of a sphere is $V = \frac{4}{3}\pi r^3$

$$2\,100 = \frac{4}{3}\pi r^3$$

$$6\,300 = 4\pi r^3$$

$$r^3 = \frac{6300}{4\pi}$$

$$r = \sqrt[3]{\frac{6300}{4\pi}}$$

$$r = 7.944\dots$$

Hence, the radius of the sphere is

7.9 cm correct to one decimal place.

7. B

$$\text{Percentage error} = \pm \frac{\text{absolute error}}{\text{measurement}} \times 100\%$$

$$\text{Absolute error} = \frac{1}{2} \times \text{precision}$$

For a measurement of 125 cm the precision is 1 cm, hence the absolute error is 0.5 cm

$$\begin{aligned} \text{Percentage error} &= \pm \frac{0.5}{125} \times 100\% \\ &= \pm 0.4\% \end{aligned}$$

8. D

Every 15° of longitude represents a time difference of 1 hour. As the difference in the angles is 75° then the difference in the time is $75^\circ \div 15^\circ = 5 \text{ hours}$

Therefore, the time of city Q could be

$$-3 - 5 = -8 \text{ or } -3 + 5 = +2$$

As -8.00 is the only option available, city Q has a Coordinated Universal Time of -8.00.

9. D

Substituting the values provided into the formula, we get

$$6 = \frac{\text{age} \times 14}{\text{age} + 12}$$

$$6(\text{age} + 12) = 14 \times \text{age}$$

$$6 \times \text{age} + 72 = 14 \times \text{age}$$

$$6 \times \text{age} - 14 \times \text{age} = -72$$

$$-8 \times \text{age} = -72$$

$$8 \times \text{age} = 72$$

$$\text{age} = \frac{72}{8} \text{ Hence, age} = 9$$

10. B

The probability of the first apple being green is $\frac{9}{24}$

If the first apple selected is green, the probability that the second will also be green is $\frac{8}{23}$

Hence, the probability of both being green is $\frac{9}{24} \times \frac{8}{23}$.

11. D

Total number of customers surveyed is

$$810 + 630 + 126 + 108 + 72 + 54 = 1800.$$

The percentage of the customers unhappy because

$$\text{of slow service is } \frac{630}{1800} \times 100 = 35\%.$$

12. B

The volume of the tank is originally 180 litres and then it falls by 8 litres every minute.

Hence $V = 180 - 8t$ which is equivalent to

$$V = -8t + 180$$

13. C

The gradient is negative and has a $\frac{\text{rise}}{\text{run}} = \frac{7}{4}$

$$\text{So } m = -\frac{7}{4} \text{ and } b = -7$$

Hence, the equation is $y = -\frac{7}{4}x - 7$.

14. A

As the distribution of the marks will all still be the same distance from the mean, the standard deviation will be the same. As all the marks are higher, the mean will be higher.

15. A

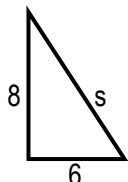
The curved area of a cone is $A = \pi r s$

We can use the curved area of the cylinder to calculate the radius using the formula $A = 2\pi r h$.

$$\text{So, } 120\pi = 2\pi \times r \times 10$$

$$\frac{120\pi}{20\pi} = r$$

$$\text{that is } r = 6 \text{ cm.}$$



$$s^2 = 8^2 + 6^2$$

$$s = \sqrt{64 + 36}$$

$$s = 10 \text{ cm.}$$

Hence, the required area is $A = \pi \times 6 \times 10$

$$A = 60\pi \text{ cm}^2$$

$$A = 188.4955 \dots \text{ cm}^2$$

$$A \approx 188 \text{ cm}^2 \text{ (3 sig. fig)}$$

SECTION II

16. The total amount of depreciation is

$$\$650 \times 3 = \$1\,950.$$

So, the purchase price was

$$\$1\,950 + \$6\,400 = \$8\,350.$$

$$17 \text{ a) } \bar{x} = \frac{750+870+632+960+642+680+982}{7}$$

$$\bar{x} = \$788$$

17 b) Using the statistics mode on the calculator.

$$\sigma = 137.687015 \dots$$

$$\sigma \approx 137.69 \text{ correct to 2 decimal places.}$$

18. 1 gram of fat contains 37 kilojoules.

This means 5.4 grams of fat contains

$$5.4 \times 37 = 199.8 \text{ kilojoules.}$$

As 1 kilocalorie = 4.184 kilojoules,

$$\text{then } 199.8 \text{ kilojoules is equal to } \frac{199.8}{4.184} = 47.75334 \dots$$

Hence, the number of kilocalories in a can of tuna containing 5.4 grams is 47.75 kcal.

19. A data distribution is described as **positively skewed** when the median and mean are greater than the mode.

20. City A is behind City B by $4 - (-8) = 12$ hours.

When it is 11:10 pm on Sunday in City B it would

be 11:10 am on Sunday in City A. Rebecca's flight will

land in City A 9 hours later at 8:10 pm on Sunday.

21. First 2 hours: \$60

$$\text{Next 5 hours: } 5 \times 1.5 \times \$28 = \$210$$

$$\text{Total amount earned: } \$270$$

22. Carol is to pay a 10% deposit on \$20 000 which is \$2 000. She must also pay $\$620 \times 12 \times 3$

$$= \$22\,320$$

So, she will pay in total

$$\$22\,320 + \$2\,000 = \$24\,320$$

So, the total interest paid is

$$I = 24\,320 - 20\,000 = \$4\,320$$

The money borrowed is

$$P = \$20\,000 - \$2\,000 = \$18\,000$$

Using the simple interest formula $I = PRN$

$$\$4\,320 = \$18\,000 \times R \times 3$$

$$\$4\,320 = \$54\,000 \times R$$

$$\text{Hence, } R = \frac{\$4\,320}{\$54\,000} \times 100 = 8\% \text{ p. a.}$$

23. $800 = 450 + 420n$

$$350 = 420n \text{ So } n = \frac{350}{420} = \frac{5}{6} \text{ hours}$$

$$= \frac{5}{6} \times 60$$

$$= 50 \text{ minutes}$$

Hence, Christina hired the helicopter for 50 minutes.

24 a) Interest $A = 20\,000 \times 0.02 = \400

$$\text{Rent } B = 560 \times 52 = \$29\,120$$

$$\text{Food bills and other. } C = \$360 \times 26 = \$9\,360$$

24 b) David's current expenses are

$$\$29\,120 + \$9\,360 + \$4\,160 = \$42\,640$$

$$\text{Increasing by 15\%} = 1.15 \times \$42\,640 = \$49\,036$$

$$\text{David's income} = \$63\,960 + \$400 = \$64\,360$$

$$\text{David could save a maximum of } \$64\,360 -$$

$$\underline{\$49\,036}$$

$$= \$15\,324$$

Hence, David's statement is correct.

25 a) As there are 17 scores, the ninth score which is 36 will be the median.

$$\text{The lower quartile will be } Q_1 = \frac{28+32}{2} = 30.$$

$$\text{The upper quartile will be } Q_3 = \frac{40+48}{2} = 44.$$

$$\text{Hence, the interquartile range is } 44 - 30 = 14.$$

25 b) 66 will be an outlier if it is greater than

$$Q_3 + 1.5 \times IQR = 44 + 1.5 \times 14 = 65.$$

Hence, 66 is an outlier

26. As $\frac{240}{150} = 1.6$ then 240 mL is 1.6 times 150 mL.

So, it will take the patient

$$\begin{aligned} 1.6 \times 1 \text{ hour } 20 \text{ minutes} &= 1.6 \times 80 \text{ minutes} \\ &= 128 \text{ minutes} \\ &= 2 \text{ hours } 8 \text{ minutes} \end{aligned}$$

27 a) A of square: $12 \times 12 = 144 \text{ cm}^2$

$$\text{A of larger quarter circle: } \frac{1}{4} \times \pi \times 8^2 = 16\pi \text{ cm}^2$$

$$\text{A of smaller quarter circle: } \frac{1}{4} \times \pi \times 4^2 = 4\pi \text{ cm}^2$$

$$\begin{aligned} \text{Therefore, A of shape} &= 144 - 16\pi - 4\pi \\ &= 144 - 20\pi \\ &= 81.17 \text{ cm}^2 \end{aligned}$$

27 b) Arc length of larger quarter circle: $\frac{1}{4} \times 2\pi \times 8$
 $= 4\pi \text{ cm}^2$

$$\begin{aligned} \text{Arc length of smaller quarter circle: } &\frac{1}{4} \times 2\pi \times 4 \\ &= 2\pi \text{ cm}^2 \end{aligned}$$

$$\text{Therefore, } P = 4 + 12 + 8 + 4\pi + 2\pi$$

$$P = 24 + 6\pi = 42.85 \text{ cm}$$

28 a) Rachel's gross income = \$ 100 400 + \$ 6 200
 $= \$ 106 600$

28 b) Tax on first \$ 87 000 = \$ 19 822

$$\begin{aligned} \text{Tax on remaining } \$ 13 400 &= \$ 13 400 \times 0.37 \\ &= \$ 4 958 \end{aligned}$$

$$\begin{aligned} \text{Total tax} &= \$ 19 822 + \$ 4 958 \\ &= \$ 24 780 \end{aligned}$$

28 c) The student may realise the new deduction will affect the amount over \$ 87 000 and so the saving would be
 $\$ 2 100 \times 0.37 = \$ 777$

Alternative method

$$\begin{aligned} \text{New taxable income} &= \$ 100 400 - \$ 2 100 \\ &= \$ 98 300 \end{aligned}$$

$$\text{Tax on first } \$ 87 000 = \$ 19 822$$

$$\begin{aligned} \text{Tax on remaining } \$ 11 300 &= \$ 11 300 \times 0.37 \\ &= \$ 4 181 \end{aligned}$$

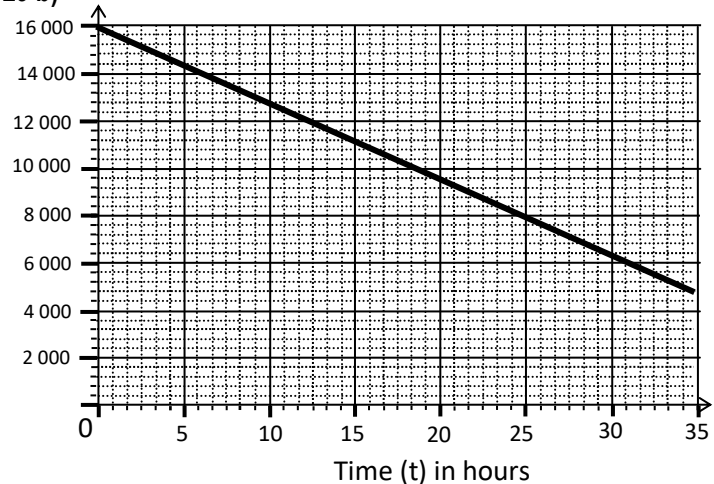
$$\begin{aligned} \text{Total tax} &= \$ 19 822 + \$ 4 181 \\ &= \$ 24 003 \end{aligned}$$

$$\begin{aligned} \text{Saving} &= \$ 24 780 - \$ 24 003 \\ &= \$ 777 \end{aligned}$$

29 a)

Time (t) in hours	0	5	10	15	20	25
Volume (V) of water in the pool in litres	16 000	14 400	12 800	11 200	9 600	8 000

29 b)



29 c) As the volume of water falls by 1600 L every 5 hours, this is equivalent to 320 L every hour.

$$\text{So } V = 16 000 - 320t$$

29 d) The pool will be emptied when $V = 0$ that is

$$0 = 16 000 - 320t$$

$$320t = 16 000$$

$$t = 50 \text{ hours}$$

Hence, it will take Donald 50 hours to empty the pool.

30. Let David's normal weekly pay be P dollars.

$$\text{So } 4P + 4P \times 0.175 = 6 345$$

$$4P + 0.7P = 6 345$$

$$4.7P = 6 345$$

$$P = \$ 1 350$$

So David's normal weekly pay is \$ 1 350.

31. From the formula sheet, $A = \frac{h}{2}(a + b)$

$$\text{So } 6 380 = \frac{x}{2}(60 + 80) + \frac{x}{2}(80 + 70)$$

$$6 380 = \frac{x}{2}(140) + \frac{x}{2}(150)$$

$$6 380 = 70x + 75x$$

$$6 380 = 145x$$

$$\text{Hence, } x = 44 \text{ m.}$$

- 32 a) To find the number of people who attended the cruise, we let $C = 3956$, we get

$$3956 = 1200 + \frac{212}{5}n$$

$$2756 = \frac{212}{5}n$$

$$65 = n$$

Hence, 65 people attended the cruise.

- 32 b) The total cost will be

$$C = 1\,200 + \frac{212}{5} \times 80$$

$$C = \$4\,592$$

Hence, the cost per person is

$$\$4592 \div 80 = \$57.40$$

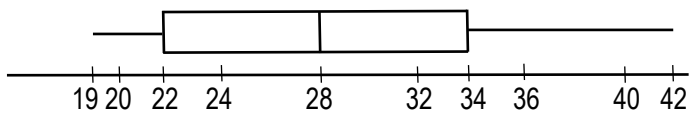
33. Using the cumulative frequency polygon:

The median will be 28

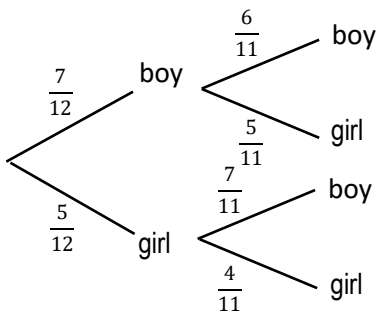
$$Q_1 = \frac{20+24}{2} = 22$$

$$Q_3 = \frac{32+36}{2} = 34$$

The lowest score is 19 and the highest is 42



- 34 a)



- 34 b) $P(G, G) = \frac{5}{12} \times \frac{4}{11}$
Hence, $P(G, G) = \frac{5}{33}$.

35. Using $T = \frac{BAC}{0.015}$
 $1.5 = \frac{BAC}{0.015}$

$$0.0225 = BAC$$

$$\text{So } 0.0225 = \frac{10 \times 3.6 - 7.5H}{6.8 \times 80}$$

$$12.24 = 36 - 7.5H$$

$$-23.76 = -7.5H$$

$$3.168 = H$$

$$3.168 \text{ hours} = 3 \text{ hours and } 10 \text{ minutes}$$

Hence, Jared must have started drinking 3 hours and 10 minutes before 12:30, that is at 9:20 pm.

36. Anthony's intake of kilocalories is
 $2 \times 162 + 180 = 504$ kilocalorie
As 1 kilocalorie = 4.184 kJ then
Anthony's intake is
 $504 \times 4.184 = 2108.736$ kJ

He uses 11.98 kJ/kg in every 30 minutes of play,
then in 75 minutes he uses

$$75 \div 30 \times 11.98 = 29.95 \text{ kJ/kg}$$

Hence, Anthony's mass is

$$M = 2108.736 \div 29.95$$

$$\approx 70.41 \text{ kg (2 dec. pl.)}$$